



The Slow Variables

京张慢变量 JINGZHANG · THE SLOW VARIABLES

How a city decides how fast AI may go

1909: the Jing-Zhang railway crossed Badaling on a switchback — steel learned to turn; today it teaches algorithms to wait.

METRICS AT A GLANCE

12.34%

green ratio

7.39%

public space

310.8k m²

building footprint

12

scenario cards

4

release gates

5

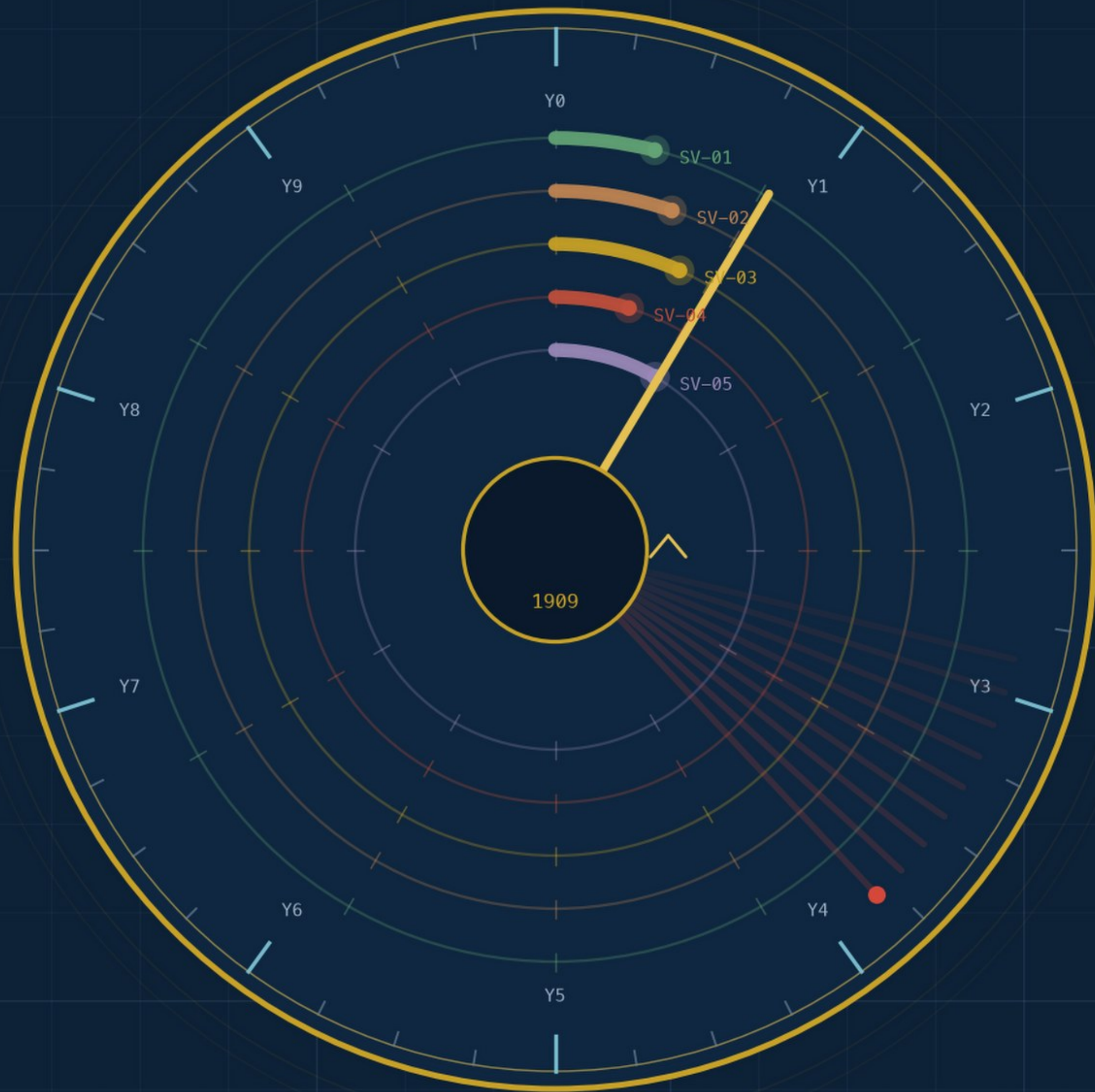
slow variables

FIG.01 总体鸟瞰 SITE OVERVIEW

FIG.01 · CENTENNIAL JING-ZHANG AI INNOVATION BELT

A city runs on two clocks

AI updates quarterly · shade, soil and daily life take years · five slow variables gate expansion



FAST clock · 1 revolution = 1 quarter

SLOW clock · 1 revolution = 10 years

CONCEPT ONLY · PROVISIONAL BOUNDARY · FIELD CHECK PENDING · NOT A STATUTORY REDLINE

THE SLOW VARIABLES 京张慢变量 · v2.0

01 Canopy

EMERGENCE · 8–15 yrs

02 Soil infiltration

EMERGENCE · 3–5 flood seasons

03 Barrier-free access

EMERGENCE · 2–3 yrs

04 Retail survival

EMERGENCE · 5–10 yrs

05 Children's independent mobility

EMERGENCE · 1–3 yrs

Test fast. Expand only slowly.

Four gates · data safety → human review → public dissent → no slow-variable decline

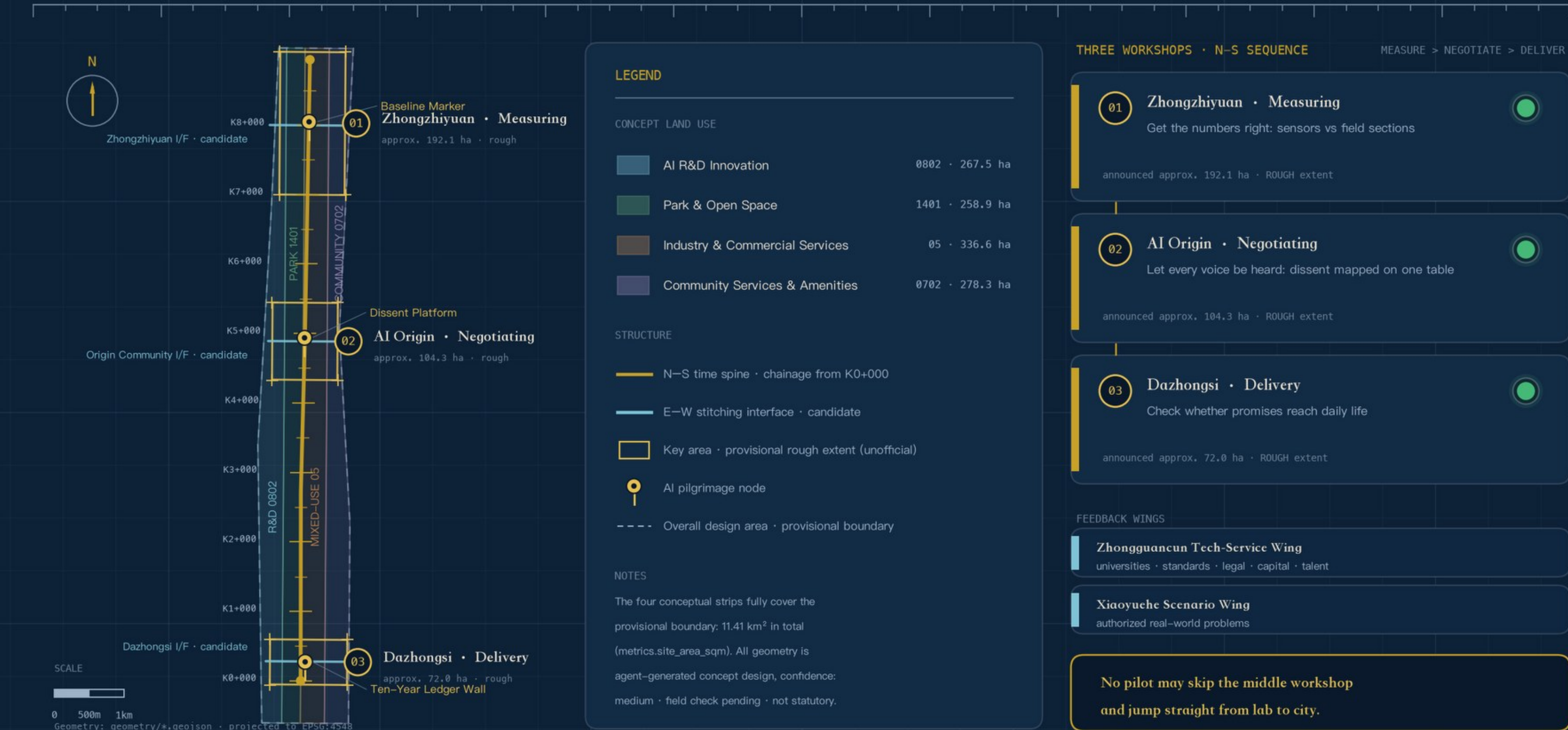
GATE 1–4

FIG.02 总体结构 STRUCTURE

FIG.02 · CENTENNIAL JING-ZHANG AI INNOVATION BELT

One Time Spine, Three Workshops

Four conceptual land-use strips cover the provisional site · a measure–negotiate–deliver sequence, with two feedback wings



两只钟 TWO CLOCKS

- FAST CLOCK · quarterly
AI updates quarterly: test, update, release
- SLOW CLOCK · by the year
five slow variables gate expansion

Experiment fast,

expansion must be reviewed slowly.

EXPERIMENT FAST · EXPAND SLOW

- | | | |
|-----------------------------------|--|--------------------------------------|
| SV-01 Canopy
8–15 yrs | SV-02 Soil infiltration
3–5 flood seasons | SV-03 Barrier-free access
2–3 yrs |
| SV-04 Retail survival
5–10 yrs | SV-05 Children's independent mobility
1–3 yrs | |

01 Zhongzhi Yard · Measure

sensors + crowd measurement, get it right

one-year baseline starts here

sensor calibration

02 AI Origin · Explain

every algorithmic decision is explainable

decisions can be questioned

explainable decisions

03 Dazhongsi · Deliver

public services delivered in daily life

the annual ledger closes here

services & annual ledger

Wings · Links · 3 Scales

three E–W links rejoin the city fabric

wings host daily life, feed data back

43.6 km² belt · 11.41 km² site · 368.4 ha key areas



Depth & Evidence

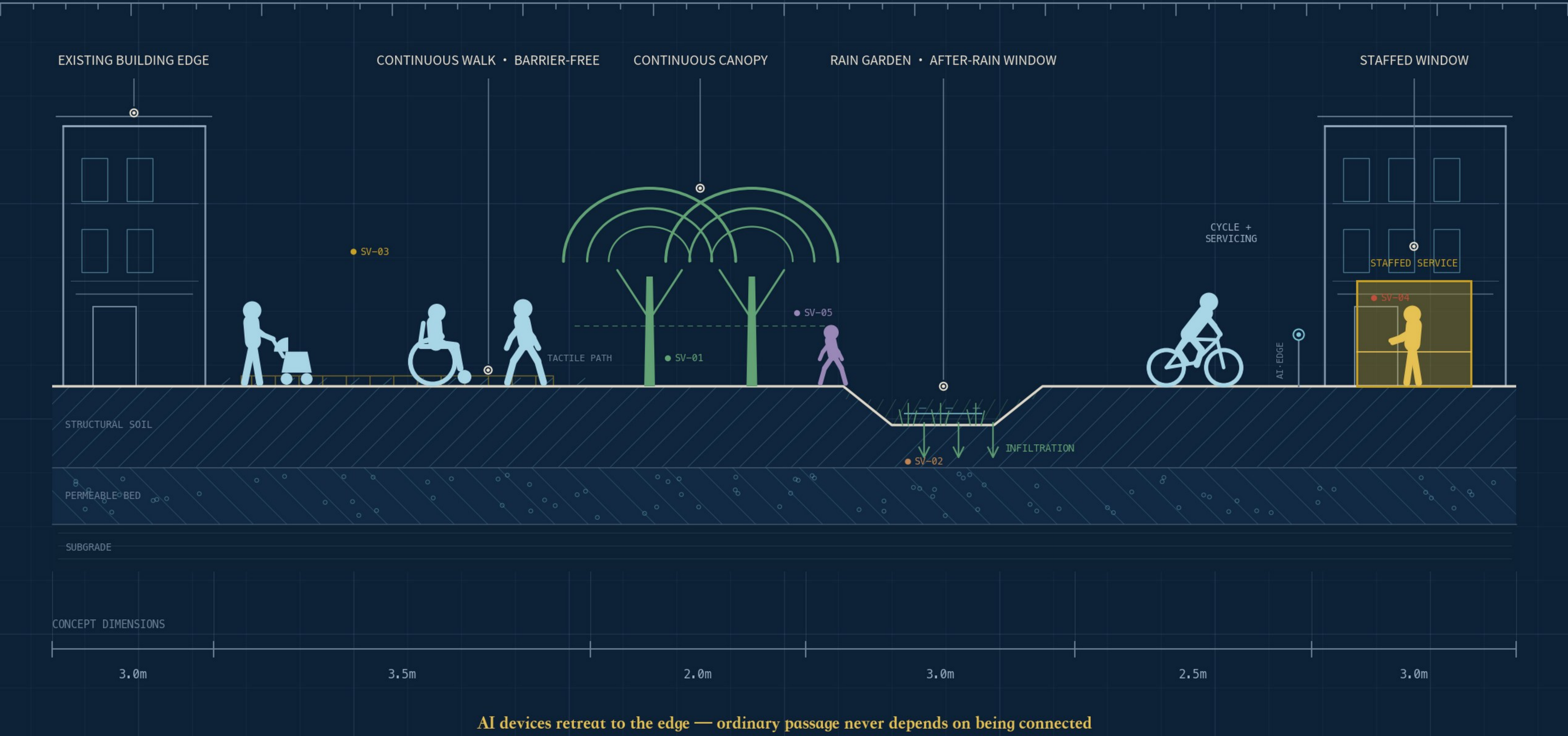
深度与证据 DEPTH & EVIDENCE

Section · key areas · mobility & blue-green · metrics

FIG.04 典型剖面 TYPICAL SECTION
FIG.04 · CENTENNIAL JING-ZHANG AI INNOVATION BELT

One street, four speeds

Typical concept section · walking, shade, infiltration and staffed service each in place



CONCEPT ONLY · PROVISIONAL BOUNDARY · FIELD CHECK PENDING · NOT A STATUTORY REDLINE

THE SLOW VARIABLES 京张慢变量 · v2.0

COMPLIANCE SNAPSHOT

PROVISIONAL

not a statutory redline

HUMAN REVIEW

signature on key decisions

DATA MINIMIZED

de-identified by default

CAN CALL A STOP

public objection channel

FOUR SCALE GATES

no gate passed, no expansion



G1

Data safety

non-compliant data, no go

DEFAULT · NO



G2

Human review

key decisions need a signature

DEFAULT · NO



G3

Public objection

residents can call a stop

DEFAULT · NO



G4

No slow-variable regression

any regression halts expansion

DEFAULT · NO

INTENTIONAL BLANKS

Blank constraint layer

reserved for statutory redlines

PENDING

Floor-area ratio

awaiting official data

PENDING

Slow-variable baseline

awaiting one-year baseline survey

Blanks are not gaps — they make room for compliance and measurement.

PENDING

FIG.03 三座工场 KEY AREAS
FIG.03 · CENTENNIAL JING-ZHANG AI INNOVATION BELT

Three workshops, one pilot

Measurements, public dissent and long-term promises cannot be settled in a single table row

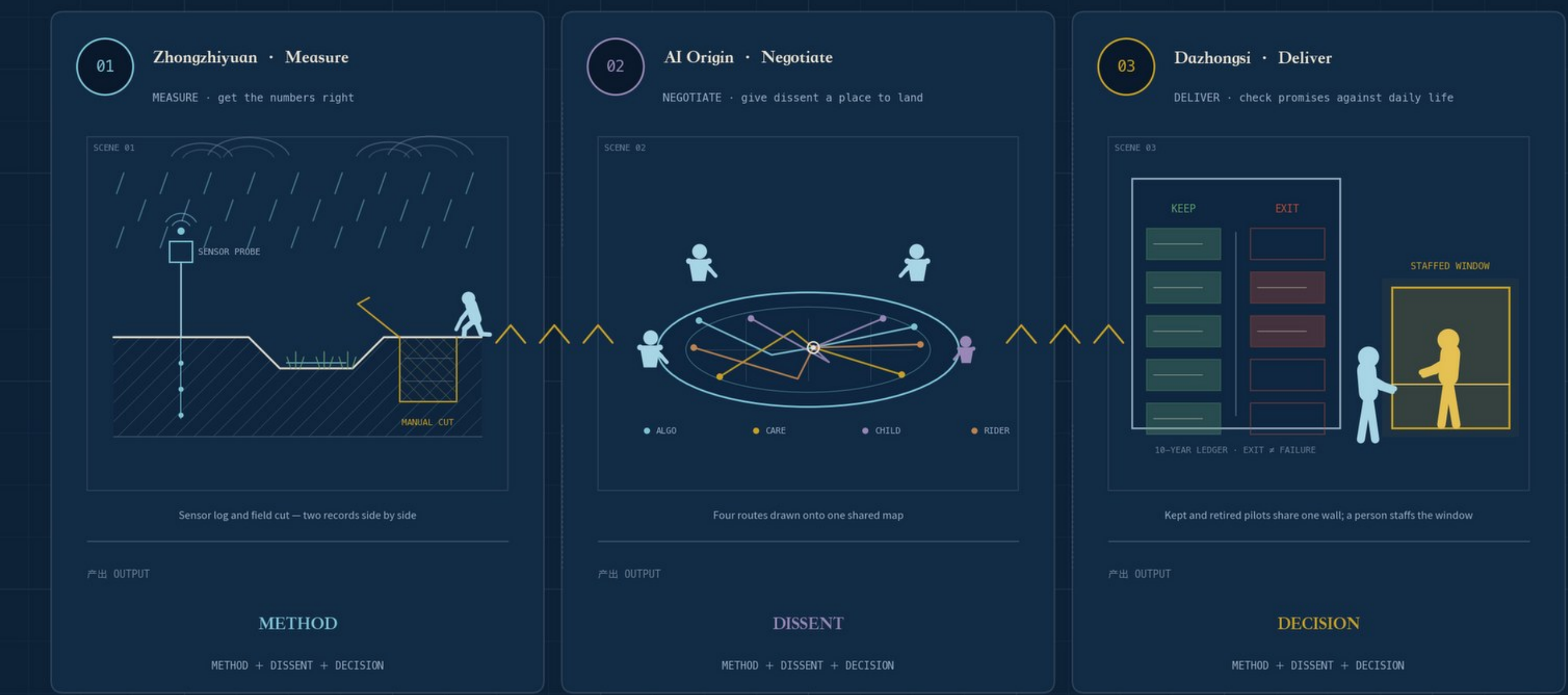


FIG.05 四种速度 MOBILITY
FIG.05 · CENTENNIAL JING-ZHANG AI INNOVATION BELT

Continuous N-S, Stitched E-W

Four everyday speeds, each with a continuous place · blue-green system takes the rain · still walkable offline

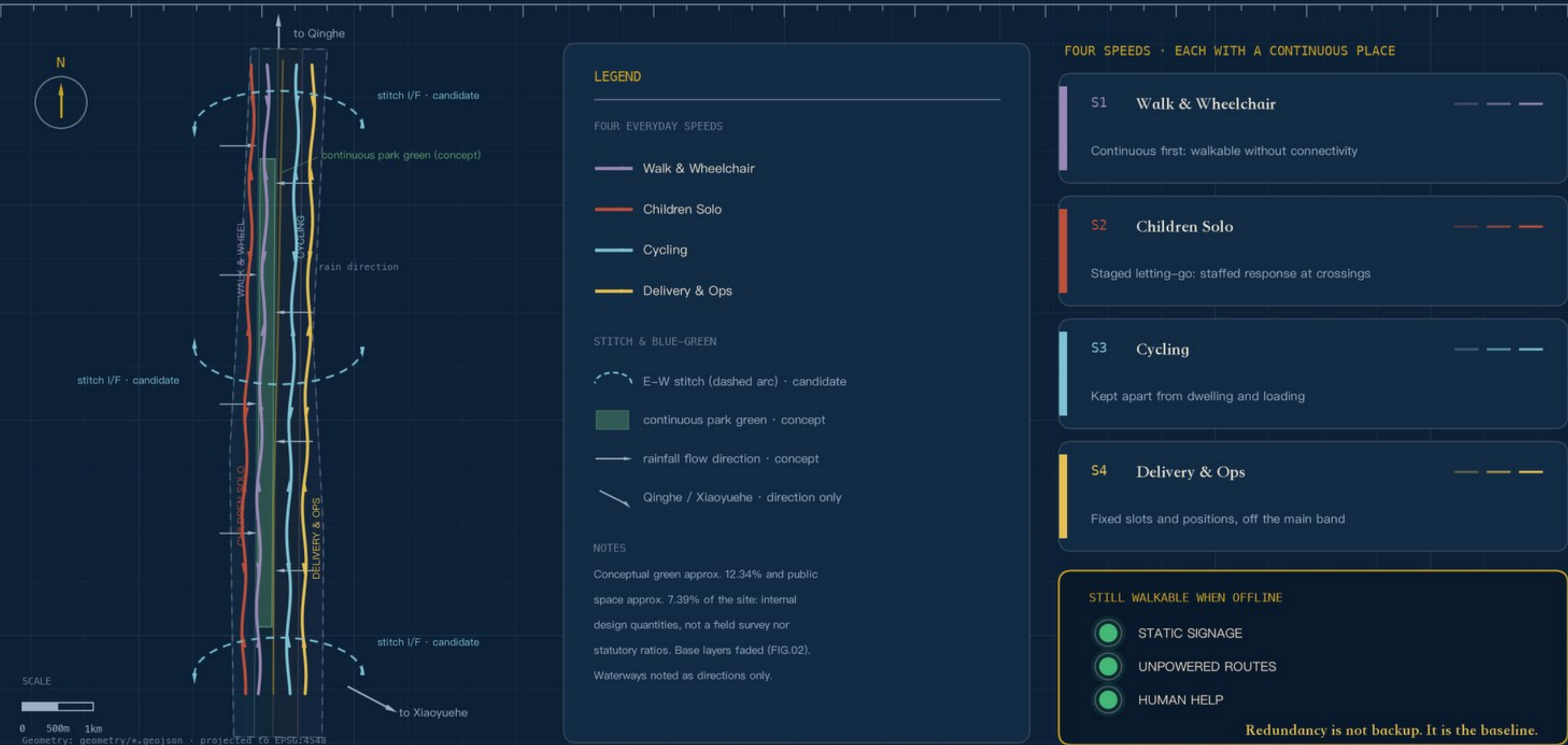


FIG.06 指标证据 METRICS
FIG.06 · CENTENNIAL JING-ZHANG AI INNOVATION BELT

Baseline first, decade goals later

Recomputable design quantities kept strictly apart from life metrics not yet known

